



MATH130: Rational Expressions and Equations

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Learning Objectives

- Find the domain of a rational expression
- Simplify rational expressions by factoring and canceling
- Solve equations involving rational expressions

Simplify each expression completely. Show all steps and circle your final answer.

Domain of a rational expression

1. Find the value(s) excluded from the domain of $f(x) = (1x + -5) / (3x + -2)$.

$$f(x) = \frac{1x + -5}{3x + -2}$$

Answer: _____

2. Find the excluded value from the domain of $g(x) = 1 / (x - 1)$.

$$g(x) = \frac{1}{x - 1}$$

Answer: _____

3. Find all values excluded from the domain of $h(x) = 3 / ((4x + 7)(2x + -6))$.

$$\frac{3}{(4x + 7)(2x + -6)}$$

Answer: _____

4. Find the value(s) excluded from the domain of $f(x) = (1x + 1) / (1x + 9)$.

$$f(x) = \frac{1x + 1}{1x + 9}$$

Answer: _____

5. Find the excluded value from the domain of $g(x) = 6 / (x - -6)$.

$$g(x) = \frac{6}{x - -6}$$

Answer: _____

6. Find all values excluded from the domain of $h(x) = 1 / ((3x + 5)(1x + 2))$.

$$\frac{1}{(3x + 5)(1x + 2)}$$

Answer: _____

7. Find the value(s) excluded from the domain of $f(x) = (4x + -1) / (3x + -2)$.

$$f(x) = \frac{4x + -1}{3x + -2}$$

Answer: _____

8. Find the excluded value from the domain of $g(x) = 4 / (x - 8)$.

$$g(x) = \frac{4}{x - 8}$$

Answer: _____

9. Find all values excluded from the domain of $h(x) = 5 / ((2x + 6)(3x + 7))$.

$$\frac{5}{(2x + 6)(3x + 7)}$$

Answer: _____

10. Find the value(s) excluded from the domain of $f(x) = (2x + 2) / (5x + 0)$.

$$f(x) = \frac{2x + 2}{5x + 0}$$

Answer: _____

11. Find the excluded value from the domain of $g(x) = 2 / (x - -2)$.

$$g(x) = \frac{2}{x - -2}$$

Answer: _____

12. Find all values excluded from the domain of $h(x) = 1 / ((3x + -8)(1x + 8))$.

$$\frac{1}{(3x + -8)(1x + 8)}$$

Answer: _____

13. Find the value(s) excluded from the domain of $f(x) = (5x + -2) / (2x + -3)$.

$$f(x) = \frac{5x + -2}{2x + -3}$$

Answer: _____

14. Find the excluded value from the domain of $g(x) = 4 / (x - -6)$.

$$g(x) = \frac{4}{x - -6}$$

Answer: _____

15. Find all values excluded from the domain of $h(x) = 4 / ((2x + -2)(2x + 1))$.

$$\frac{4}{(2x + -2)(2x + 1)}$$

Answer: _____

16. Find the value(s) excluded from the domain of $f(x) = (5x + -3) / (1x + -2)$.

$$f(x) = \frac{5x + -3}{1x + -2}$$

Answer: _____

17. Find the excluded value from the domain of $g(x) = 8 / (x - 6)$.

$$g(x) = \frac{8}{x - 6}$$

Answer: _____

18. Find all values excluded from the domain of $h(x) = 4 / ((3x + 1)(3x + -1))$.

$$\frac{4}{(3x + 1)(3x + -1)}$$

Answer: _____

19. Find the value(s) excluded from the domain of $f(x) = (1x + -1) / (4x + -2)$.

$$f(x) = \frac{1x + -1}{4x + -2}$$

Answer: _____

20. Find the excluded value from the domain of $g(x) = 2 / (x - -5)$.

$$g(x) = \frac{2}{x - -5}$$

Answer: _____

21. Find all values excluded from the domain of $h(x) = 6 / ((1x + -5)(1x + -6))$.

$$\frac{6}{(1x + -5)(1x + -6)}$$

Answer: _____

22. Find the value(s) excluded from the domain of $f(x) = (1x + -1) / (4x + 5)$.

$$f(x) = \frac{1x + -1}{4x + 5}$$

Answer: _____

23. Find the excluded value from the domain of $g(x) = 6 / (x - 8)$.

$$g(x) = \frac{6}{x - 8}$$

Answer: _____

Simplifying rational expressions

24. Simplify the rational expression: $(4x + 20) / (4x + 20)$. Find the excluded value and the simplified form.

$$\frac{4x + 20}{4x + 20}$$

Answer: _____

25. Simplify the rational expression: $(3x + 6) / (3x + 15)$. Find the excluded value and the simplified form.

$$\frac{3x + 6}{3x + 15}$$

Answer: _____

26. Simplify the rational expression: $(3x + 3) / (3x + 18)$. Find the excluded value and the simplified form.

$$\frac{3x + 3}{3x + 18}$$

Answer: _____

27. Simplify the rational expression: $(1x + 4) / (1x + 6)$. Find the excluded value and the simplified form.

$$\frac{1x + 4}{1x + 6}$$

Answer: _____

28. Simplify the rational expression: $(3x + 9) / (3x + 15)$. Find the excluded value and the simplified form.

$$\frac{3x + 9}{3x + 15}$$

Answer: _____

29. Simplify the rational expression: $(4x + 4) / (4x + 24)$. Find the excluded value and the simplified form.

$$\frac{4x + 4}{4x + 24}$$

Answer: _____

30. Simplify the rational expression: $(4x + 24) / (4x + 24)$. Find the excluded value and the simplified form.

$$\frac{4x + 24}{4x + 24}$$

Answer: _____



MATH130: Rational Expressions and Equations

ANSWER KEY & SOLUTIONS

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Topics: Simplifying rational expressions, Domain of a rational expression. All answers verified by independent computation.

Solutions

Domain of a rational expression

1. Find the value(s) excluded from the domain of $f(x) = (1x + -5) / (3x + -2)$.

$$f(x) = \frac{1x + -5}{3x + -2}$$

→ Set denominator = 0: $3x + -2 = 0 \Rightarrow x = 2/3$.

→ The domain excludes $x = 2/3$.

Answer: $x \neq 2/3$

2. Find the excluded value from the domain of $g(x) = 1 / (x - 1)$.

$$g(x) = \frac{1}{x - 1}$$

→ Set $x - 1 = 0 \Rightarrow x = 1$.

→ Domain: all real numbers except $x = 1$.

Answer: $x \neq 1$

3. Find all values excluded from the domain of $h(x) = 3 / ((4x + 7)(2x + -6))$.

$$\frac{3}{(4x + 7)(2x + -6)}$$

→ Set each factor to zero.

→ $4x + 7 = 0 \Rightarrow x = -7/4$.

→ $2x + -6 = 0 \Rightarrow x = 3$.

→ Excluded values: $x = -7/4$ and $x = 3$.

Answer: $x \neq -7/4, x \neq 3$

4. Find the value(s) excluded from the domain of $f(x) = (1x + 1) / (1x + 9)$.

$$f(x) = \frac{1x + 1}{1x + 9}$$

→ Set denominator = 0: $1x + 9 = 0 \Rightarrow x = -9$.

→ The domain excludes $x = -9$.

Answer: $x \neq -9$

5. Find the excluded value from the domain of $g(x) = 6 / (x - -6)$.

$$g(x) = \frac{6}{x - -6}$$

→ Set $x - -6 = 0 \Rightarrow x = -6$.

→ Domain: all real numbers except $x = -6$.

Answer: $x \neq -6$

6. Find all values excluded from the domain of $h(x) = 1 / ((3x + 5)(1x + 2))$.

$$\frac{1}{(3x + 5)(1x + 2)}$$

→ Set each factor to zero.

$$\rightarrow 3x + 5 = 0 \Rightarrow x = -5/3.$$

$$\rightarrow 1x + 2 = 0 \Rightarrow x = -2.$$

→ Excluded values: $x = -5/3$ and $x = -2$.

Answer: $x \neq -5/3, x \neq -2$

7. Find the value(s) excluded from the domain of $f(x) = (4x + -1) / (3x + -2)$.

$$f(x) = \frac{4x + -1}{3x + -2}$$

→ Set denominator = 0: $3x + -2 = 0 \Rightarrow x = 2/3$.

→ The domain excludes $x = 2/3$.

Answer: $x \neq 2/3$

8. Find the excluded value from the domain of $g(x) = 4 / (x - 8)$.

$$g(x) = \frac{4}{x - 8}$$

→ Set $x - 8 = 0 \Rightarrow x = 8$.

→ Domain: all real numbers except $x = 8$.

Answer: $x \neq 8$

9. Find all values excluded from the domain of $h(x) = 5 / ((2x + 6)(3x + 7))$.

$$\frac{5}{(2x + 6)(3x + 7)}$$

→ Set each factor to zero.

$$\rightarrow 2x + 6 = 0 \Rightarrow x = -3.$$

$$\rightarrow 3x + 7 = 0 \Rightarrow x = -7/3.$$

→ Excluded values: $x = -3$ and $x = -7/3$.

Answer: $x \neq -3, x \neq -7/3$

10. Find the value(s) excluded from the domain of $f(x) = (2x + 2) / (5x + 0)$.

$$f(x) = \frac{2x + 2}{5x + 0}$$

→ Set denominator = 0: $5x + 0 = 0 \Rightarrow x = 0$.

→ The domain excludes $x = 0$.

Answer: $x \neq 0$

11. Find the excluded value from the domain of $g(x) = 2 / (x - -2)$.

$$g(x) = \frac{2}{x - -2}$$

→ Set $x - -2 = 0 \Rightarrow x = -2$.

→ Domain: all real numbers except $x = -2$.

Answer: $x \neq -2$

12. Find all values excluded from the domain of $h(x) = 1 / ((3x + -8)(1x + 8))$.

$$\frac{1}{(3x + -8)(1x + 8)}$$

→ Set each factor to zero.

$$\rightarrow 3x + -8 = 0 \Rightarrow x = 8/3.$$

$$\rightarrow 1x + 8 = 0 \Rightarrow x = -8.$$

→ Excluded values: $x = 8/3$ and $x = -8$.

Answer: $x \neq 8/3, x \neq -8$

13. Find the value(s) excluded from the domain of $f(x) = (5x + -2) / (2x + -3)$.

$$f(x) = \frac{5x + -2}{2x + -3}$$

→ Set denominator = 0: $2x + -3 = 0 \Rightarrow x = 3/2$.

→ The domain excludes $x = 3/2$.

Answer: $x \neq 3/2$

14. Find the excluded value from the domain of $g(x) = 4 / (x - -6)$.

$$g(x) = \frac{4}{x - -6}$$

→ Set $x - -6 = 0 \Rightarrow x = -6$.

→ Domain: all real numbers except $x = -6$.

Answer: $x \neq -6$

15. Find all values excluded from the domain of $h(x) = 4 / ((2x + -2)(2x + 1))$.

$$\frac{4}{(2x + -2)(2x + 1)}$$

→ Set each factor to zero.

$$\rightarrow 2x + -2 = 0 \Rightarrow x = 1.$$

$$\rightarrow 2x + 1 = 0 \Rightarrow x = -1/2.$$

→ Excluded values: $x = 1$ and $x = -1/2$.

Answer: $x \neq 1, x \neq -1/2$

16. Find the value(s) excluded from the domain of $f(x) = (5x + -3) / (1x + -2)$.

$$f(x) = \frac{5x + -3}{1x + -2}$$

→ Set denominator = 0: $1x + -2 = 0 \Rightarrow x = 2$.

→ The domain excludes $x = 2$.

Answer: $x \neq 2$

17. Find the excluded value from the domain of $g(x) = 8 / (x - 6)$.

$$g(x) = \frac{8}{x - 6}$$

→ Set $x - 6 = 0 \Rightarrow x = 6$.

→ Domain: all real numbers except $x = 6$.

Answer: $x \neq 6$

18. Find all values excluded from the domain of $h(x) = 4 / ((3x + 1)(3x + -1))$.

$$\frac{4}{(3x + 1)(3x + -1)}$$

→ Set each factor to zero.

$$\rightarrow 3x + 1 = 0 \Rightarrow x = -1/3.$$

$$\rightarrow 3x + -1 = 0 \Rightarrow x = 1/3.$$

→ Excluded values: $x = -1/3$ and $x = 1/3$.

Answer: $x \neq -1/3, x \neq 1/3$

19. Find the value(s) excluded from the domain of $f(x) = (1x + -1) / (4x + -2)$.

$$f(x) = \frac{1x + -1}{4x + -2}$$

→ Set denominator = 0: $4x + -2 = 0 \Rightarrow x = 1/2$.

→ The domain excludes $x = 1/2$.

Answer: $x \neq 1/2$

20. Find the excluded value from the domain of $g(x) = 2 / (x - -5)$.

$$g(x) = \frac{2}{x - -5}$$

→ Set $x - -5 = 0 \Rightarrow x = -5$.

→ Domain: all real numbers except $x = -5$.

Answer: $x \neq -5$

21. Find all values excluded from the domain of $h(x) = 6 / ((1x + -5)(1x + -6))$.

$$\frac{6}{(1x + -5)(1x + -6)}$$

→ Set each factor to zero.

$$\rightarrow 1x + -5 = 0 \Rightarrow x = 5.$$

$$\rightarrow 1x + -6 = 0 \Rightarrow x = 6.$$

→ Excluded values: $x = 5$ and $x = 6$.

Answer: $x \neq 5, x \neq 6$

22. Find the value(s) excluded from the domain of $f(x) = (1x + -1) / (4x + 5)$.

$$f(x) = \frac{1x + -1}{4x + 5}$$

→ Set denominator = 0: $4x + 5 = 0 \Rightarrow x = -5/4$.

→ The domain excludes $x = -5/4$.

Answer: $x \neq -5/4$

23. Find the excluded value from the domain of $g(x) = 6 / (x - 8)$.

$$g(x) = \frac{6}{x - 8}$$

→ Set $x - 8 = 0 \Rightarrow x = 8$.

→ Domain: all real numbers except $x = 8$.

Answer: $x \neq 8$

Simplifying rational expressions

24. Simplify the rational expression: $(4x + 20) / (4x + 20)$. Find the excluded value and the simplified form.

$$\frac{4x + 20}{4x + 20}$$

→ Factor: $4(x + 5) / 4(x + 5)$.

→ Cancel the common factor 4: $(x + 5) / (x + 5)$.

→ Excluded value: $x = -5$ (denominator = 0).

Answer: $\frac{x + 5}{x + 5}, x \neq -5$

25. Simplify the rational expression: $(3x + 6) / (3x + 15)$. Find the excluded value and the simplified form.

$$\frac{3x + 6}{3x + 15}$$

→ Factor: $3(x + 2) / 3(x + 5)$.

→ Cancel the common factor 3: $(x + 2) / (x + 5)$.

→ Excluded value: $x = -5$ (denominator = 0).

Answer: $\frac{x + 2}{x + 5}, x \neq -5$

26. Simplify the rational expression: $(3x + 3) / (3x + 18)$. Find the excluded value and the simplified form.

$$\frac{3x + 3}{3x + 18}$$

→ Factor: $3(x + 1) / 3(x + 6)$.

→ Cancel the common factor 3: $(x + 1) / (x + 6)$.

→ Excluded value: $x = -6$ (denominator = 0).

Answer: $\frac{x + 1}{x + 6}, x \neq -6$

27. Simplify the rational expression: $(1x + 4) / (1x + 6)$. Find the excluded value and the simplified form.

$$\frac{1x + 4}{1x + 6}$$

→ Factor: $1(x + 4) / 1(x + 6)$.

→ Cancel the common factor 1: $(x + 4) / (x + 6)$.

→ Excluded value: $x = -6$ (denominator = 0).

Answer: $\frac{x + 4}{x + 6}, x \neq -6$

28. Simplify the rational expression: $(3x + 9) / (3x + 15)$. Find the excluded value and the simplified form.

$$\frac{3x + 9}{3x + 15}$$

→ Factor: $3(x + 3) / 3(x + 5)$.

→ Cancel the common factor 3: $(x + 3) / (x + 5)$.

→ Excluded value: $x = -5$ (denominator = 0).

Answer: $\frac{x + 3}{x + 5}, x \neq -5$

29. Simplify the rational expression: $(4x + 4) / (4x + 24)$. Find the excluded value and the simplified form.

$$\frac{4x + 4}{4x + 24}$$

→ Factor: $4(x + 1) / 4(x + 6)$.

→ Cancel the common factor 4: $(x + 1) / (x + 6)$.

→ Excluded value: $x = -6$ (denominator = 0).

Answer: $\frac{x + 1}{x + 6}, x \neq -6$

30. Simplify the rational expression: $(4x + 24) / (4x + 24)$. Find the excluded value and the simplified form.

$$\frac{4x + 24}{4x + 24}$$

→ Factor: $4(x + 6) / 4(x + 6)$.

→ Cancel the common factor 4: $(x + 6) / (x + 6)$.

→ Excluded value: $x = -6$ (denominator = 0).

Answer: $\frac{x + 6}{x + 6}, x \neq -6$
